

Saral-KYC: Simple-Secure-Seamless

Concept Overview

Saral-KYC reimagines KYC as an *AI-first, compliance-aware orchestration system* that automates end-to-end onboarding while preserving transparency, fairness, and auditability. It unifies **multimodal AI**, **large language models (LLMs)**, and **risk intelligence graphs** to deliver a *trustworthy, scalable, and explainable KYC pipeline* for banks and financial institutions.

Core System Architecture & Automation Design

The Saral-KYC architecture operates as a **multi-layered AI orchestration pipeline** combining document intelligence, entity analytics, and compliance reasoning – optimized for both *accuracy* and *regulatory transparency*.

1. Document Intelligence & Data Extraction:

Incoming documents (Aadhar, PAN, utility bills, etc.) pass through a *multimodal preprocessing pipeline* that detects type, orientation, and authenticity before extraction.

- **Vision Transformers (Donut / DocFormer):** Perform layout-aware parsing of IDs, extracting key-value pairs directly from images.
- **OCR + NER Hybrid Layer:** Handles non-standard document layouts using OCR text fed to a transformer-based *Named Entity Recognition* model fine-tuned for Indian KYC fields.
- **Forgery & Liveness Verification:** CNNs detect tampering (compression artifacts, edge mismatches) while 3D face depth estimation confirms selfie liveness and cross-matches with document photos.
- **Cross-Document Validation:** Sentence-BERT embeddings compare extracted entities across documents, identifying inconsistencies in names, addresses, or ID details.

2. AI-driven Risk & Anomaly Detection:

Verified entities feed into a *Risk Intelligence Graph*, mapping relationships between users, addresses, and connected accounts.

- **Graph Neural Networks (GNNs):** Detect relational anomalies such as synthetic identities or fraud clusters.
- **LLM-based Behavior Understanding:** Analyzes free-text statements or user chats to infer intent, cooperation, and risk sentiment.

- **Dynamic Risk Scoring Engine:** Combines document authenticity, entity consistency, and behavioral trust into a composite *KYC Risk Index*, aligned with FATF/AML guidelines. Scoring rules are version-controlled for audit readiness.

3. Explainability, Governance & Compliance Integration:

Each model decision generates an *XAI metadata packet* containing SHAP explanations, decision rationale, and applied regulatory rule IDs.

- A *Regulatory Logic Engine* dynamically aligns actions with RBI/AML policies via rule-sync APIs.
- Continuous *model monitoring* tracks fairness, accuracy, and drift – auto-triggering retraining pipelines when thresholds are breached.
- An *immutable audit ledger* logs every decision, ensuring end-to-end traceability.

4. Human Oversight & Feedback Loop:

Ambiguous cases are escalated with AI-generated rationale summaries for manual verification. Human verdicts reinforce retraining datasets, enabling a *self-improving governance framework*.

Innovation

- **Conversational LLM Orchestrator:** Guides customers contextually, providing multilingual clarifications and dynamic document assistance.
 - **Adaptive Fairness Checks:** Differential performance monitoring ensures unbiased decisions across user segments.
 - **Real-time Reg-Tech Sync:** Policy updates auto-reflect in decision logic through modular rule APIs.
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Impact

- **Efficiency:** 70% reduction in manual verification time and near-instant approval for low-risk profiles.
- **Compliance:** Full explainability and auditable decisions reduce regulatory risk.
- **Scalability:** Cloud-native microservices integrate easily with legacy banking systems.